

Alexander Hernandez

Portfolio Project

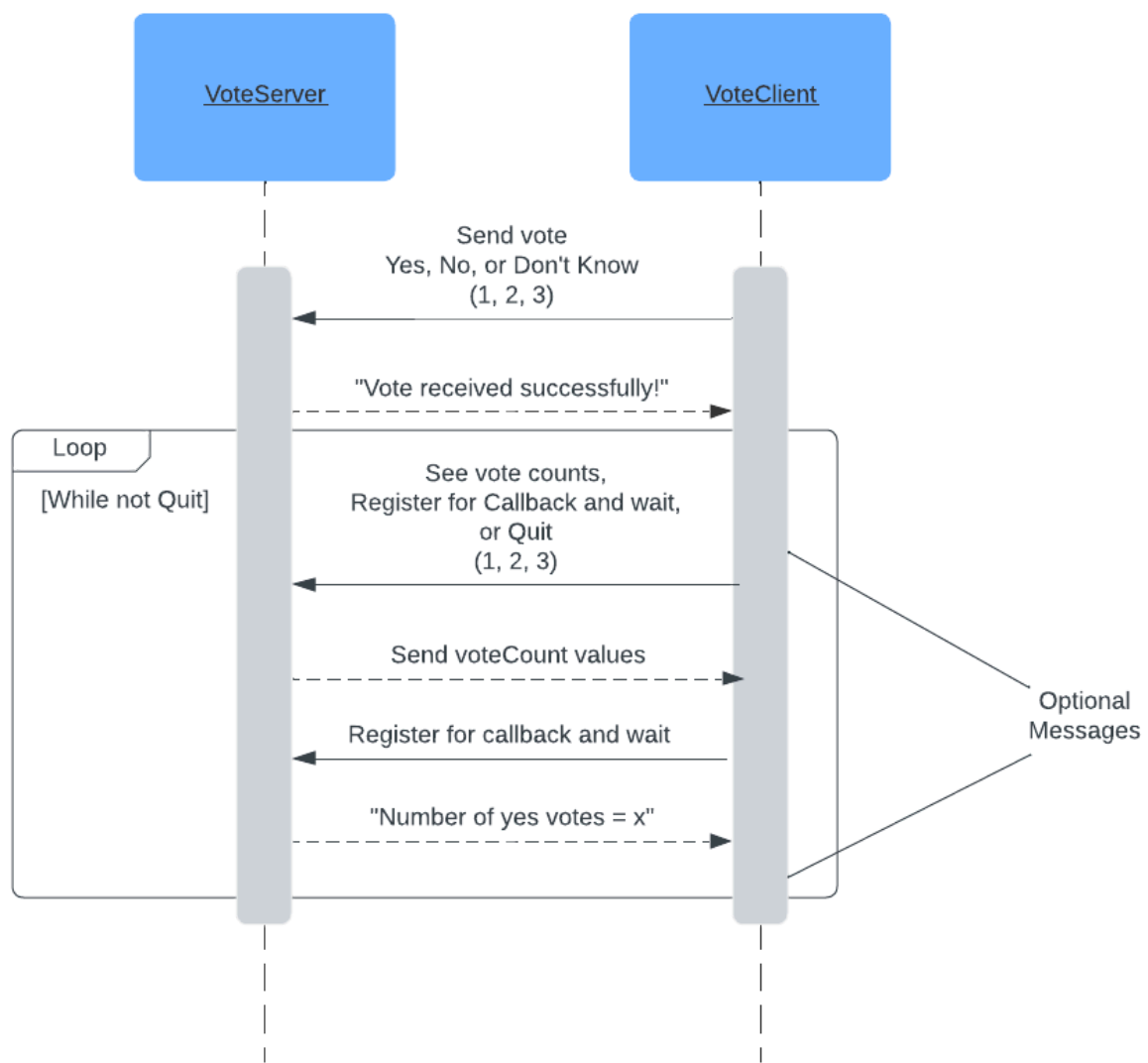
4/10/2024

Portfolio Project Report

Source Code:

Source Code/Project located within the zip. Instructions for running are in the README.

Sequence Diagram:



Detailed Functions of Application:

1. **VoteClient class:**
 - Runs the VoteClient. Has functionality to prompt user, send messages, create callback objects, and output to user.
2. **VoteServer class:**
 - Runs the VoteServer.
3. **VoteClientInterface interface:**
 - Defines methods for client communication: notifyMe, getMessage, getClientId, isNotificationReceived.
4. **VoteClientImpl class:**
 - Implements VoteClientInterface methods for client communication.
5. **VoteServerInterface interface:**
 - Defines methods for server communication: sendVote, getVoteCounts, registerCallback, unregisterCallback.
6. **VoteServerImpl class:**
 - Implements VoteServerInterface methods for server functionality and callback management.
7. **VoteGUI class:**
 - Provides a GUI for creating clients and leading into the VoteWindow. VoteGUI maintains the same functionality of the VoteClient.
8. **VoteWindow class:**
 - Provides a GUI for voting functionality, registering for callbacks, and getting vote counts.

Testing Results:

VoteClient and VoteServer Testing -

Startup - .bat file successfully runs

```
C:\> Select C:\WINDOWS\system32\cmd.exe  
-Running Vote clients and server-  
Waiting for 2 seconds, press a key to continue ...
```

VoteClient1 - Vote Selection and Viewing Vote Counts

```
C:\Program Files\Common Files\Oracle\Java\javapath\java.exe  
Cast your vote by entering the corresponding number:  
1. Yes  
2. No  
3. Don't Know  
Enter your choice: 1  
Vote received successfully!  
Do you want to:  
1. See vote counts  
2. Register for callback and wait  
3. Quit  
Enter your choice: 1  
Vote counts:  
Yes: 1, No: 0, Don't Care: 0
```

VoteClient2 - Vote Selection and Registration for Callback

```
Do you want to:  
1. See vote counts  
2. Register for callback and wait  
3. Quit  
Enter your choice: 2  
Registered for callback. Waiting for notification...
```

VoteClient - Vote Selection and Quit (Successful, instantly closes)

```
C:\Program Files\Common Files\Oracle\Java\javapath\java.exe
Cast your vote by entering the corresponding number:
1. Yes
2. No
3. Don't Know
Enter your choice: 3
Vote received successfully!
Do you want to:
1. See vote counts
2. Register for callback and wait
3. Quit
Enter your choice: 3
```

VoteServer:

- 1) Registry Creation – Successful
- 2) Registered Object for Callback – Successful
- 3) ERROR on Callback Unregistration – Unsuccessful (Bonus implementation)

```
C:\Program Files\Common Files\Oracle\Java\javapath\java.exe
RMI registry cannot be located at port 1234
RMI registry created at port 1234
Vote Server ready.
Registered callback object: vote.VoteClientImpl@e6dc1475
Unregister callback error - client wasn't registered: vote.VoteClientImpl@e6dc1475
Unregister callback error - client wasn't registered: vote.VoteClientImpl@e6dc1475
Unregister callback error - client wasn't registered: vote.VoteClientImpl@65ea17fc
```

VoteServer – Callbacks with 1 Callback Successful

```
Registered callback object: vote.VoteClientImpl@aba5254c
*****
Server initiating callbacks ---
doing 0-th callback

Sending message - Received Callback message: Number of yes votes = 2
*****
Server completed callbacks ---
```

VoteGUI –

Startup - .bat file successfully runs

 C:\WINDOWS\system32\cmd.exe

```
-Running Vote GUI with server-
```

```
Waiting for 2 seconds, press a key to continue ...
```

VoteGUI - Set Up Host and Port

 Vote Client   

- Host

localhost

- Port

1234

Connect

VoteGUI – Vote Received Successfully

 Vote   

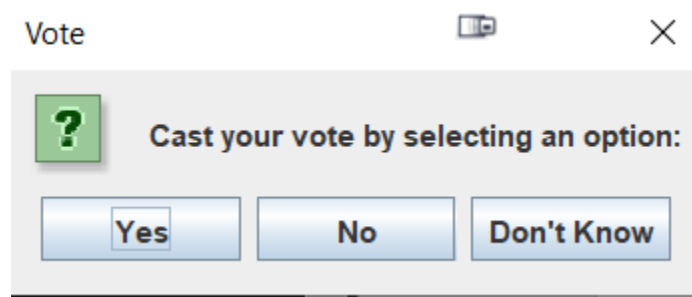
Vote

Register for Callback

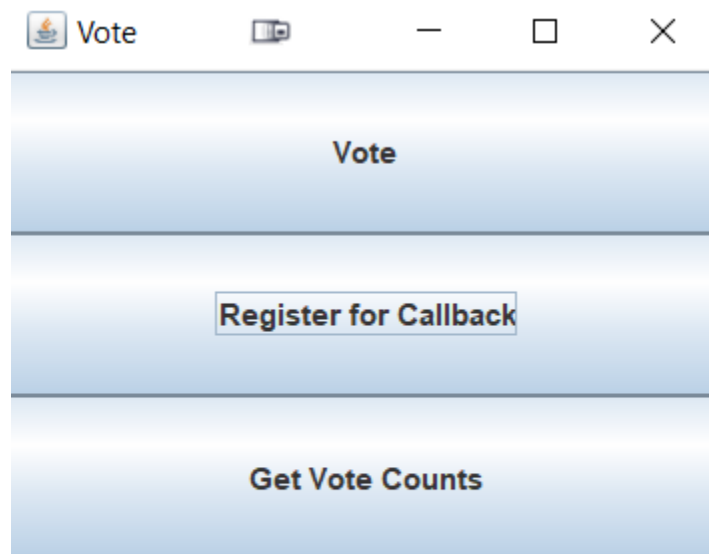
Get Vote Counts

Vote received successfully!

VoteGUI – Vote Casting

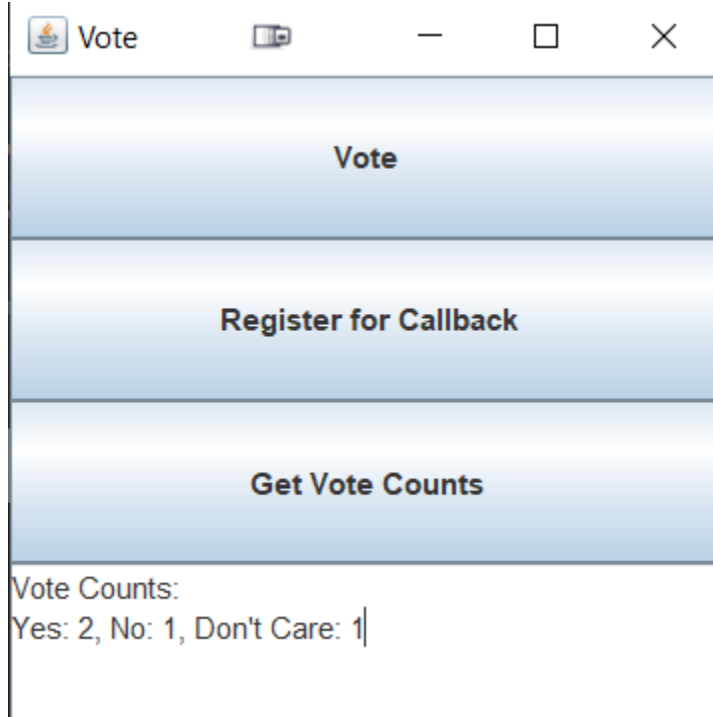


VoteGUI – Registered for Callback Successfully



Registered for callback. Waiting for notification...

VoteGUI – Get Vote Counts Successful



VoteServer – Successful same as VoteClient and VoteServer

C:\Program Files\Common Files\Oracle\Java\javapath\java.exe

```
RMI registry cannot be located at port 1234
RMI registry created at port 1234
Vote Server ready.
Registered callback object: vote.VoteClientImpl@5dd9aade
*****
Server initiating callbacks ---
doing 0-th callback

Sending message - Received Callback message: Number of yes votes = 2
*****
Server completed callbacks ---
```